

Problem to Incident

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Chapter 1. Purpose

This document is a **prescriptive reference implementation** for the **Log to Incident** use case: when an application emits an **ERROR** or **WARN** log line, observability detects it, ServiceNow Event Management ingests an event, an alert is raised on the correct infrastructure CI, and an incident is opened on the correct CSDM Application Service — using the **Spark Service-Side Log Pattern** (Kubernetes pods) or the **Spark Client-Side Log Pattern** (enterprise PySpark drivers).

The flow is vendor-specific only where the platforms require it (Dynatrace OpenPipeline, Service Graph Connector, CSDM). Application names, host names, and tenant identifiers in examples reflect the **brooks-lab Spark** reference implementation unless noted otherwise. Implementors may automate these steps with Ansible, Terraform, REST APIs, or manual configuration.

External references (vendor documentation):

- [Dynatrace — Log Analytics and OpenPipeline](#)
- [Davis problem detection](#)
- [ServiceNow — Event Management](#)

1.1. Terminology: SGC vs SGO-Dynatrace

SGC (Service Graph Connector) is the ServiceNow connector product family. **SGO** is **Service Graph Observability** — the observability-specific variant delivered by the Store app **Service Graph Connector for Observability – Dynatrace** (`sn_dynatrace_integ`).

SGO-Dynatrace is the **literal platform identifier**, not a typo for SGC. ServiceNow uses it as:

- the `source=` query parameter on the Event Management inbound webhook URL;
- `sys_object_source.name` and CMDB `discovery_source` for imported Dynatrace entities;
- the filter value in business rules and EM rules (`source=SGO-Dynatrace`).

Do **not** rename it to **SGC-Dynatrace** in configuration — the scoped app, existing CMDB rows, and webhook URLs expect the exact string **SGO-Dynatrace**.

1.2. Naming conventions (ServiceNow and Dynatrace specifications)

Prescriptive automation in this document **must** use technology- or pattern-specific names, not workload-specific names (for example **Spark**).

Category	Rule
ServiceNow Script Includes	Name by pattern , not application: <code>K8sLogPodCiBind</code> , <code>ResolveApplicationService</code> — not <code>SparkLogPodCiBind</code>

Category	Rule
ServiceNow business rules	Name by table + pattern : <code>em-event-bind-k8s-log-pod-ci</code> , <code>em-alert-create-k8s-log-incident</code>
Dynatrace OpenPipeline attributes	Use <code>k8s.pod_name</code> , <code>k8s.log.path</code> — not <code>spark.pod_name</code> (lab path values in matchers may still reflect deployment layout)
Davis <code>event.name</code>	Use Application log {loglevel} on {k8s.pod_name} — workload examples (Spark) belong in scenario prose only
ServiceNow <code>message_key</code> prefix	Use <code>K8sLog-{pod}-</code> for per-pod deduplication
Incident short description	Use Event Log WARN / Event Log ERROR — not <code>Spark Event Log ...</code> or generic <code>Log Errors</code>
Lab examples	Host names, pod names, CSDM identifiers (<code>spark-master</code>), and log paths (<code>/mnt/spark/logs/...</code>) may appear in scenario tables and figures as illustrative values only

Implementors deploying for a non-Kubernetes workload should substitute the appropriate pattern (for example Docker process group binding) while keeping the same separation: generic automation names, illustrative example values.

Chapter 2. Log to Incident

2.1. Overview

Log to Incident supports **two incident-mapping patterns** that share the same Dynatrace OpenPipeline → Davis → ServiceNow Event Management pipeline (Steps 2–4) but diverge at **CMDB prerequisites** (Step 0), **log emission** (Step 1), and **incident correlation** (Step 5):

- **Spark Service-Side Log Pattern** — Kubernetes pod logs; `incident.cmdb_ci` via **Contains** traversal from pod CI to Application Service.
- **Spark Client-Side Log Pattern** — PySpark driver logs on any host; `incident.cmdb_ci` via **log path contract** to a manual logical Application Service.

Use the comparison table below to see where each pattern applies and how incidents resolve. Detailed CMDB models and correlation algorithms appear in [CMDB model](#) and [Incident correlation](#).

Pattern	Where the workload runs	Incident CI resolution
Service-side	Kubernetes pods (Spark master, worker, history server, executors)	Spark Service-Side Log Pattern — traverse Contains::Contained by from <code>cmdb_ci_kubernetes_pod</code> to Application Service
Client-side	PySpark driver JVMs on any host (lab servers today; enterprise laptops and VMs tomorrow)	Spark Client-Side Log Pattern — fixed path segment <code>/logs/spark-client/</code> → Spark Client Application Service (<code>service_mapping: manual</code>)

Both patterns require the **full formatted log line** in event, alert, and incident descriptions; Dynatrace **ERROR** or **WARN** detection; and `incident.cmdb_ci = cmdb_ci_service_discovered` (Application Service), not the infrastructure CI on the alert.

2.2. End-to-end flow

Both patterns follow the same observability path from application log line to ServiceNow incident. Figure 1 is the navigation map; pattern-specific configuration is called out in Steps 0, 1, and 5.

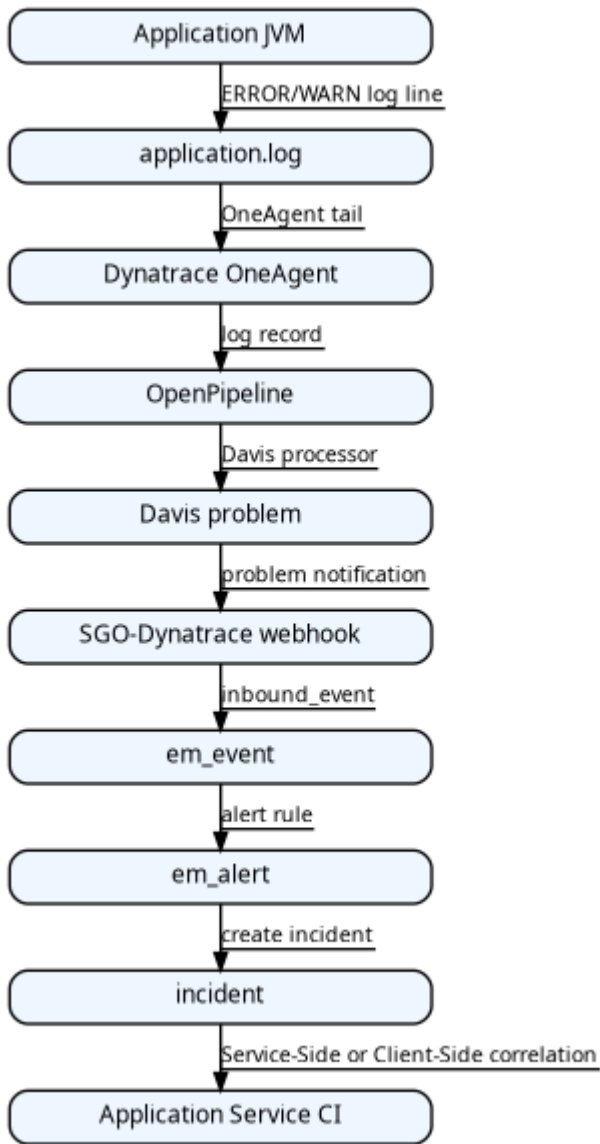


Figure 1. Log to Incident end-to-end data flow

The step table maps each layer to the section that configures it. Steps 2–4 are shared; Steps 0, 1, and 5 include parallel **Service-side** and **Client-side** subsections.

Step	Layer	Outcome
0	ServiceNow CMDB / SGC	Application Services, workload CIs, tags, <code>sys_object_source</code> ; pattern-specific CMDB model (<code>CMDB model</code>)
1	Application / Log4j	ERROR/WARN on a path that encodes workload identity (pod name or <code>spark-client</code>)
2	Dynatrace detection	OpenPipeline matches WARN/ERROR
3	Dynatrace → ServiceNow	Webhook carries log line and impacted entity
4	ServiceNow Event Management	<code>em_event</code> / <code>em_alert</code> bind to infrastructure CI
5	ServiceNow ITSM	<code>incident.cmdb_ci</code> = Application Service (<code>Service-side pattern</code> or <code>Client-side pattern</code>)

2.3. Step 0 — CMDB, CSDM, and SGC prerequisites

Before log problems can bind to infrastructure CIs and incidents to an **Application Service**, ServiceNow must contain the CSDM hierarchy, pattern-specific CMDB models ([CMDB model](#)), and incident automation ([Incident correlation](#)).

2.3.1. ServiceNow — CSDM Application Services

Declare one Application Service per logical service boundary. Each Spark Kubernetes workload that should roll up to that service carries a matching pod label synced to `cmdb_key_value`.

The table maps brooks-lab Spark Application Service identifiers to pod roles and the label each pod must carry.

Application Service identifier	Workload role	Tag on pod (label key=value)
spark-master	Spark master StatefulSet	servicenow.io/application-service-identifier=spark-master
spark-history-server	Spark history server	servicenow.io/application-service-identifier=spark-history-server
spark-worker-lab1	Spark worker on Lab1	servicenow.io/application-service-identifier=spark-worker-lab1
spark-worker-lab2	Spark worker on Lab2	servicenow.io/application-service-identifier=spark-worker-lab2

Reference CSDM fragment from `servicenow/regions/brooks-lab/spark.csdm.yaml`:

```
application_services:
- name: Spark Master
  identifier: spark-master
  short_description: Spark master – Kubernetes StatefulSet
  operational_status: "1"
  parent_business_service: Apache Spark
  platform: kubernetes
  service_mapping: tags
  discover: false
  service_tier: app
  environment: on-prem
  location: brooks-lab
  cluster: brooks-lab
  namespace: spark
  owned_by: gbrooks

- name: Spark Worker (Lab1)
  identifier: spark-worker-lab1
  short_description: Spark worker on Lab1
  platform: kubernetes
  service_mapping: tags
  discover: false
  namespace: spark
```

CSDM deployment must create:

- cmdb_ci_service_discovered rows and Business Application → Business Service → Application Service **Contains** hierarchy
- Declared **depends_on** edges (shared infrastructure the service depends on)
- tag_list on each tag-based Application Service (Service Mapping Operations REST) so the mapping engine knows which cmdb_key_value labels qualify

2.3.2. ServiceNow — Kubernetes pod CIs and tags

Pod manifest labels are the single source of truth for tag-based Service Mapping on Spark workloads.

Activity	Produces
Kubernetes discovery / KVA resync	cmdb_ci_kubernetes_pod rows for running pods
Label sync to CMDB (REST upsert on pod CI)	cmdb_key_value rows for servicenow.io/* labels copied from pod metadata.labels

Pod manifest labels (brooks-lab Spark worker example — apply at deploy time):

```
apiVersion: v1
kind: Pod
metadata:
  name: spark-worker-lab1-0
  labels:
    servicenow.io/application-service-identifier: spark-worker-lab1
    servicenow.io/environment: on-prem
    servicenow.io/location: brooks-lab
    app.kubernetes.io/name: spark-worker
spec:
  containers:
    - name: spark-worker
      env:
        - name: POD_NAME
          valueFrom:
            fieldRef:
              fieldPath: metadata.name
      volumeMounts:
        - name: app-logs
          mountPath: /opt/spark/logs
          subPathExpr: ${POD_NAME}
  volumes:
    - name: app-logs
      persistentVolumeClaim:
        claimName: spark-logs-pvc
```

After tags exist on pod CIs, run **tag-based Service Mapping** (scheduled job or **Update map** in Service Mapping Workspace) to materialize:

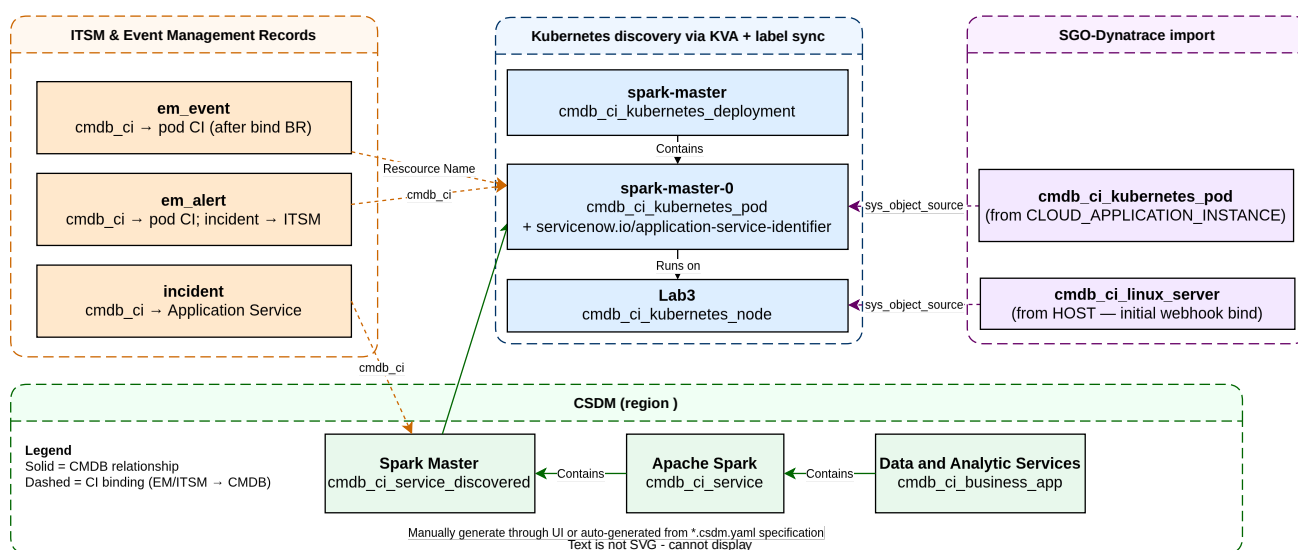
cmdb_ci_service_discovered → **Contains::Contained by** → cmdb_ci_kubernetes_pod

2.3.3. CMDB model

Alerts bind to **infrastructure** CIs; incidents bind to **Application Services**. Each Log to Incident pattern requires a different CMDB shape so operators assign the most informative CI at each layer. The subsections below define **Service-side** and **Client-side** models side by side.

2.3.3.1. Service-side

The Service-side model shows logical relationships between CSDM Application Services, Kubernetes-discovered CIs, SGC-imported entities, and Event Management bindings for pod-scoped logs. Use it to verify **em_alert.cmdb_ci** → pod and **incident.cmdb_ci** → Application Service before enabling automation.



Lab scenario: WARN from **Master:** logger in `/mnt/spark/logs/spark-master-0/spark-app.log`. ServiceNow binds **em_event.cmdb_ci** / **em_alert.cmdb_ci** to **cmdb_ci_kubernetes_pod**. **incident.cmdb_ci** is **cmdb_ci_service_discovered** for Application Service **spark-master** (from `servicenow/regions/brooks-lab/spark.csdm.yaml`), via the **Spark Service-Side Log Pattern** — not the Dynatrace problem object and not the pod CI.

The table below lists each CMDB object in the Service-side scenario and which Event Management / ITSM field it populates.

Object	Role in this scenario
cmdb_ci_business_app — Data and Analytic Services	CSDM parent; not an incident CI
cmdb_ci_service — Apache Spark	Business Service parent of Application Service
cmdb_ci_service_discovered — spark-master	incident.cmdb_ci (preferred)
cmdb_ci_kubernetes_pod — spark-master-0	em_event.cmdb_ci , em_alert.cmdb_ci ; resource = <code>/mnt/spark/logs/spark-master-0/spark-app.log</code>
cmdb_ci_linux_server — Lab3	Fallback when pod entity lookup fails; Runs on target for pod

Object	Role in this scenario
CLOUD_APPLICATION_INSTANCE → cmdb_ci_kubernetes_pod	Dynatrace entity merged via <code>sys_object_source</code>

Tag-based Application Services (`service_mapping: tags, platform: kubernetes`) **must** materialize **Contains::Contained** by edges from Application Service to member pods **before** the first qualifying log line.

Design note: Davis `event.name` should scope problems (lab: `Application log {loglevel} on {k8s.pod_name}`). ServiceNow `message_key` should include the **pod name** (`K8sLog-spark-master-0-...`) so unrelated pods are not correlated into one bucket.

2.3.3.2. Client-side

The Client-side model shows a **host-independent** logical Application Service for PySpark drivers that may run on workstations outside CMDB discovery scope. There is **no required Contains edge** from client hosts to the Application Service.

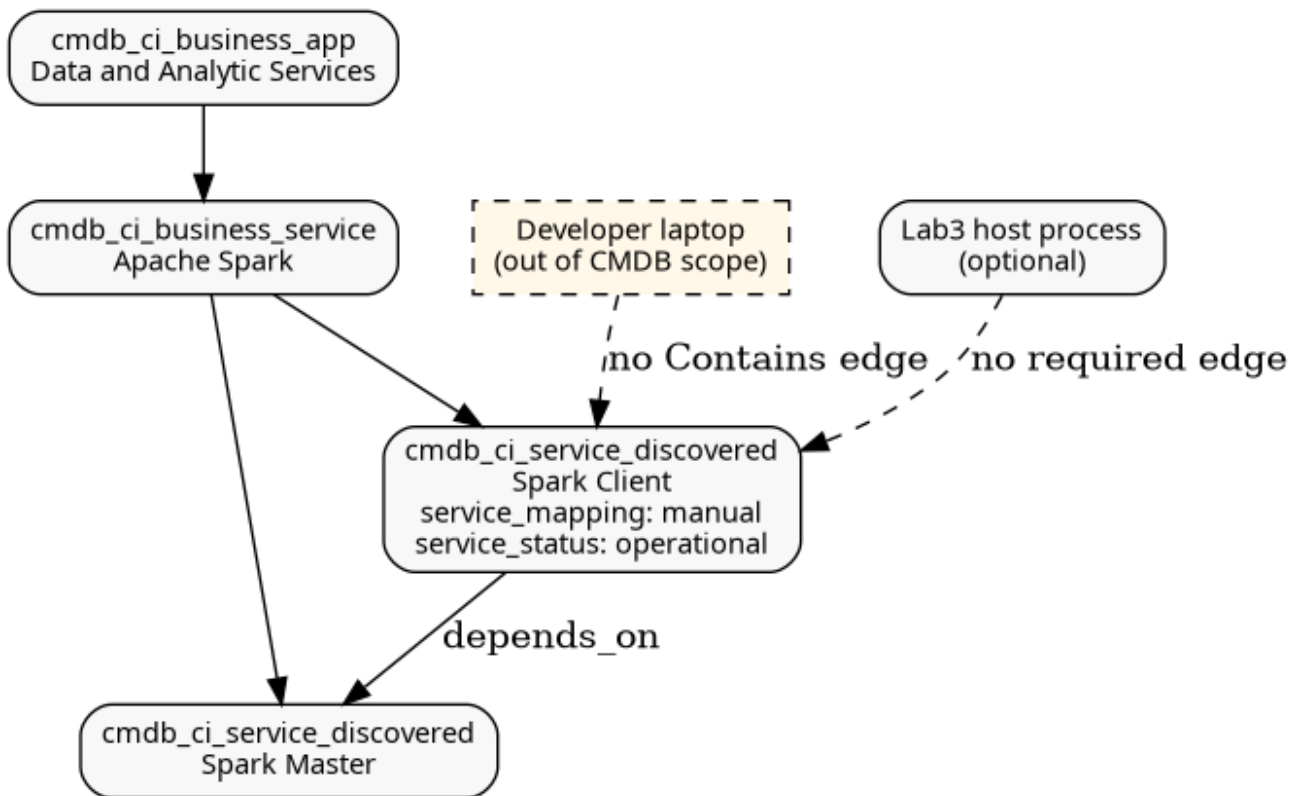


Figure 2. Enterprise client CMDB model

Use the table below to see which objects participate in Client-side alerts and incidents.

Object	Role
cmdb_ci_service_discovered — Spark Client	incident.cmdb_ci (path-based lookup); no required Contains children
cmdb_ci_linux_server — client host	May appear on <code>em_alert.cmdb_ci</code> when Davis binds to HOST; not used for incident CI

Object	Role
Per-driver <instance> segment	Disambiguates concurrent drivers; not used for AS lookup
Optional cmdb_ci_appl on in-scope hosts	Optional enrichment when OneAgent + SGC import a tagged process; not required for incidents

CSDM fragment (brooks-lab):

```
- name: Spark Client
  identifier: spark-client
  platform: host
  service_mapping: manual
  discover: false
  depends_on:
    - Spark Master
```

2.3.4. ServiceNow — SGC topology import and sys_object_source

SGC scheduled import	Dynatrace entity type	CMDB class	Role in log incidents
Hosts	HOST	cmdb_ci_linux_server	Fallback when log tails on host and pod entity unavailable
Kubernetes Pod	CLOUD_APPLICATION_INSTANCE	cmdb_ci_kubernetes_pod	Preferred alert/event CI for pod-scoped logs
Process Groups	PROCESS_GROUP	cmdb_ci_appl	Alternative when Davis binds to JVM process
Application Relationships	Smartscape edges	cmdb_rel_ci	Optional topology context

Each imported entity receives a **sys_object_source** row: name=SGO-Dynatrace, id=<Dynatrace entityId>, target_sys_id → CMDB CI sys_id.

Configure on the ServiceNow instance:

- Service Graph Connector for Observability – Dynatrace (scoped app sn_dynatrace_integ)
- Inbound event listener: POST /api/sn_em_connector/em/inbound_event?source=SGO-Dynatrace
- Scheduled imports for hosts, Kubernetes pods, and process groups covering the cluster namespace

2.3.5. Dynatrace — management zone

Place hosts, Kubernetes cluster entities, and pods that run the application in a **management zone** dedicated to the integration (for example **Application Observability**). The **alerting profile**

for ServiceNow problem notifications must scope to that management zone so only in-scope problems are forwarded.

2.3.6. Step 0 completion criteria

Complete the shared criteria first, then the pattern-specific rows for each Log to Incident pattern you deploy.

#	Criterion (shared)
1	SGC imported hosts, Kubernetes pods, and process groups; <code>sys_object_source</code> rows exist for entity IDs in the management zone
2	Dynatrace management zone scopes hosts and pods that emit qualifying logs
3	Event Management rules mark SGO-Dynatrace log events Ready when <code>cmdb_ci</code> is populated (Event Management rules)
4	Incident business rules deployed: pod bind (K8s log pod bind business rules), create incident (Incident correlation)

2.3.6.1. Service-side

#	Criterion
1	Application Service CIs exist for each K8s Spark role; <code>tag_list</code> configured for tag-based mapping
2	Pod labels synced to <code>cmdb_key_value</code> on <code>cmdb_ci_kubernetes_pod</code>
3	Service Mapping materialized Contains from Application Service to member pods
4	Incident automation implements Service-Side Contains traversal (Service-side pattern)

2.3.6.2. Client-side

#	Criterion
1	Application Service Spark Client exists with <code>service_mapping: manual</code> and <code>service_status=operational</code>
2	Client drivers write under <code>/mnt/spark/logs/spark-client/<instance>/</code> (Client-side log emission)
3	Dynatrace custom log source and OpenPipeline list spark-client paths before generic <code>/mnt/spark/logs/*/</code>
4	Incident automation implements <code>resolveFromSparkClientLogPath()</code> (Client-side pattern)

2.4. Step 1 — Application emits a log line with requisite metadata

2.4.1. Required log content

Attribute	Source	Consumed by
Timestamp, level, logger, line number, message	Log4j PatternLayout (or equivalent)	Dynatrace content field; ServiceNow em_event.description
Workload directory in file path	Kubernetes subPathExpr: \$(POD_NAME) on log volume	OpenPipeline path parser → pod name
Shared storage path	PVC or hostPath mount visible to OneAgent	Dynatrace custom log source pattern

2.4.2. Log4j configuration (reference)

Use a **RollingFile** appender only (no console appender in production pods). Set **APP_LOG_DIR** or **SPARK_LOG_DIR** to the pod-specific directory.

Key fields: **fileName** and **filePattern** path prefixes **must** match the Dynatrace custom log source paths in [custom log source specification](#) and the OpenPipeline **log.source** matchers in [OpenPipeline specification](#) (brooks-lab: `/mnt/spark/logs/<pod>/spark-app.log` and rotated `spark-app-*.log`).

Specification 2.4.2-1. Log4j RollingFile (Spark client / chapter reference)

```
# RollingFileAppender – file logging
appender.rolling.type = RollingFile
appender.rolling.name = RollingFile
appender.rolling.fileName = ${env:SPARK_LOG_DIR:-/mnt/spark/logs/client}/spark-app.log
appender.rolling.filePattern = ${env:SPARK_LOG_DIR:-/mnt/spark/logs/client}/spark-app-%d{yyyy-MM-dd}-%i.log
appender.rolling.layout.type = PatternLayout
appender.rolling.layout.pattern = %d{yyyy-MM-dd HH:mm:ss} %-5p %c{1}:%L - %m%n%ex
```

Pass the configuration to the JVM:

```
-Dlog4j2.configurationFile=file:///opt/myapp/conf/log4j2.properties
```

Optional: add Mapped Diagnostic Context (MDC) for pod name in the log line itself (redundant when the path encodes pod identity, but useful if log.source is normalized to a wildcard pattern):

```
appender.rolling.layout.pattern = %d{yyyy-MM-dd HH:mm:ss} %-5p [%X{podName}] %c{1}:%L - %m%n%ex
```

Set MDC at process startup from the **POD_NAME** environment variable.

2.4.3. Log emission

Log emission defines where application JVMs write ERROR/WARN lines so OneAgent can tail them and OpenPipeline can extract workload identity. Requirements diverge by pattern; configure **Service-side** and **Client-side** subsections as applicable.

2.4.3.1. Service-side

Service-side workloads **must** run as **Kubernetes pods** with pod-scoped log directories visible on

the node. The requirement table lists each constraint and why it matters for path-based pod binding.

Requirement	Specification
Runtime	Workload runs as a Kubernetes pod (not a host-local client driver)
Pod identity	<code>metadata.name</code> is the directory name in the log file path
Log4j appender	RollingFile only in production pods; no console-only logging for incident-bound workloads
Log directory	<code>subPathExpr: \$(POD_NAME)</code> on a volume mounted at a path OneAgent can tail on the node (brooks-lab: <code>/mnt/spark/logs/\$(POD_NAME)/</code>)
Log file name	<code>application.log</code> or rotated <code>application-.log</code> (brooks-lab Spark: <code>spark-app.log</code> , <code>spark-app-.log</code>)
Pod label	<code>servicenow.io/application-service-identifier=<identifier></code> matching a CSDM Application Service (Step 0)
Levels	Log4j emits ERROR and WARN for conditions that should open incidents per severity policy

The init container or `subPathExpr` mount must ensure each pod writes to a unique directory whose name equals `metadata.name`:

```
env:
- name: APP_LOG_DIR
  value: /var/log/myapp/$(POD_NAME)
- name: POD_NAME
  valueFrom:
    fieldRef:
      fieldPath: metadata.name
volumeMounts:
- name: app-logs
  mountPath: /var/log/myapp
  subPathExpr: $(POD_NAME)
```

Resulting path on the node (OneAgent tail source):

```
/mnt/spark/logs/myapp-worker-7d4f9b-xk2lm/application.log
└─ must equal cmdb_ci_kubernetes_pod.name
```

2.4.3.2. Client-side

The **Spark Client-Side Log Pattern** requires a fixed path segment (`spark-client`) and one subdirectory per driver JVM. The attribute table maps each path element to the consumer that resolves incidents.

Attribute	Source	Consumed by
Timestamp, level, logger, message	Log4j PatternLayout	Dynatrace <code>loglevel</code> , <code>content</code> ; ServiceNow <code>em_event.description</code>

Attribute	Source	Consumed by
Fixed AS path segment <code>spark-client</code>	<code>run-chapters.sh</code> sets <code>SPARK_LOG_DIR=/mnt/spark/logs/spark-client/<instance></code>	OpenPipeline Davis matcher; incident BR (Client-side pattern)
Per-driver instance directory	<code>{hostname}-{pid}</code> (override with <code>SPARK_DRIVER_INSTANCE</code>)	OneAgent enrichment → <code>spark.client.instance</code>
Optional process tag	<code>DT_TAGS=servicenow.io/application-service-identifier=spark-client</code>	Process-group problems only; not required for log-path incidents

Example layout:

```
/mnt/spark/logs/spark-client/lab3-12345/spark-app.log
/mnt/spark/logs/spark-client/lab3-12345/spark-app-2026-07-06-1.log
```

Do not colocate multiple concurrent drivers in one directory — Log4j RollingFile would interleave or truncate `spark-app.log`.

2.4.4. Log path and OneAgent tail (not NFS)

Dynatrace **does not read logs from the NFS server**. OneAgent on each **Kubernetes node** tails log files on the **node's local filesystem**:

1. Inside the pod, Log4j writes to a pod-local mount (for example `/opt/spark/logs/${POD_NAME}/spark-app.log`).
2. A **hostPath** (or equivalent node-visible mount) maps that directory to `/mnt/spark/logs/${POD_NAME}/` on the node where the pod is scheduled.
3. OneAgent **custom log source** `LOG_PATH_PATTERN` matches `/mnt/spark/logs/*/...` on the **host** and tails those files into Grail.
4. NFS volumes used for shared Spark data (`/mnt/spark/events`, `/mnt/spark/data`, ...) are **not** the Dynatrace log tail source.

2.4.5. Dynatrace and ServiceNow (Step 1)

No Dynatrace or ServiceNow configuration at log emission time. **Preconditions:** OneAgent can read the log path on the node; the path directory name equals `cmdb_ci_kubernetes_pod.name` after KVA discovery; pod carries `servicenow.io/application-service-identifier`; tag-based Service Mapping has materialized **Contains::Contained by** from Application Service to pod **before** the first qualifying log line (Step 0).

2.5. Step 2 — Dynatrace detects ERROR and WARN log lines

Logs ingested through **OpenPipeline** are evaluated by pipeline processors. Classic

builtin:logmonitoring.log-events rules do **not** apply to logs routed through OpenPipeline.

2.5.1. OpenPipeline processor flow



Figure 3. OpenPipeline log alerts processor chain

2.5.2. Dynatrace objects to configure

Object	Settings schema	Purpose
Custom log source	builtin:logmonitoring.custom-log-source-settings	OneAgent tails application log paths on hosts or nodes
Logs routing	builtin:openpipeline.logs.routing	Route matched logs into the log-alerts pipeline
Log alerts pipeline	builtin:openpipeline.logs.pipelines	Extract pod name, set dt.source_entity, Davis processor
Alerting profile	builtin:alerting.profile	Forward in-scope problems to ServiceNow notification
Problem notification	builtin:problem.notifications	POST webhook on problem OPEN and RESOLVED

2.5.3. Custom log source

Register path patterns that cover per-workload directories and rotated files.

Key fields: **path** patterns **must** align with Log4j **fileName** / **filePattern** ([Log4j RollingFile specification](#)) and OpenPipeline **log.source** matchers ([OpenPipeline specification](#)). **Pod** paths use segment = `cmdb_ci_kubernetes_pod.name`. **Client-mode** paths use `/mnt/spark/logs/spark-client/<instance>/` ([Client-side pattern](#)) with enrichment `spark.client.as_identifier` and `spark.client.instance`. OpenPipeline **drops** legacy **-chapter** paths parsed as pod names. **enrichment** attributes `k8s.log.path` (`${0}`) and `k8s.pod_name` or `spark.client.instance` (`${1}`) **must** be set on wildcard paths – list **more specific** patterns (`spark-client`) **before** generic `/mnt/spark/logs/`.

Specification 2.5.1-1. Custom log source (brooks-lab Spark)

```
{
  "schemaId": "builtin:logmonitoring.custom-log-source-settings",
  "scope": "environment",
  "value": {
    "enabled": true,
    "config-item-title": "Spark Lab - application log files",
    "custom-log-source": {
      "type": "LOG_PATH_PATTERN",
      "values-and-enrichment": [
        {
          "path": "/mnt/spark/logs/*/spark-app.log",
          "enrichment": [
            { "type": "attribute", "key": "k8s.log.path", "value": "${0}" },

```



```

    { "type": "attribute", "key": "k8s.pod_name", "value": "${1}" }
  ]
},
{
  "path": "/mnt/spark/logs/*/spark-app-*.log",
  "enrichment": [
    { "type": "attribute", "key": "k8s.log.path", "value": "${0}" },
    { "type": "attribute", "key": "k8s.pod_name", "value": "${1}" }
  ]
},
{ "path": "/opt/spark/logs/spark-app.log*" }
]
}
}
}

```

2.5.4. Logs routing

Route OneAgent-sourced records whose log.source matches the application path into the alerts pipeline:

```

{
  "schemaId": "builtin:openpipeline.logs.routing",
  "scope": "environment",
  "value": {
    "routing": [
      {
        "enabled": true,
        "description": "Application ERROR/WARN log alerts",
        "matcher": "matchesValue(log.source, \"/var/log/myapp/*\")",
        "pipelines": ["application-log-alerts"]
      }
    ]
  }
}

```

2.5.5. Log alerts pipeline (complete reference)

The pipeline has three processor stages before storage:

1. **Fields extract** — parse **k8s.pod_name** from **log.source** when the path is literal.
2. **Fields add** / **entity lookup** — set **dt.source_entity** via **lookupEntity(CLOUD_APPLICATION_INSTANCE, k8s.pod_name)**.
3. **Davis processor** — match **WARN/ERROR**; set **event.name** to include pod name; emit **event.description** with full log line.
4. **Bucket assignment** — store in **default_logs** bucket.

Key fields: DQL **matcher** on **log.source**; **lookupEntity**; Davis **event.name**, **event.description**, **dt.source_entity**.

Specification 2.5.4-1. OpenPipeline log alerts pipeline (brooks-lab Spark)

```

{
  "schemaId": "builtin:openpipeline.logs.pipelines",
  "scope": "environment",
  "value": {

```

```

"displayName": "Spark Lab - log alerts",
"customId": "spark-lab-log-alerts",
"processing": {
  "processors": [
    {
      "id": "extract-spark-pod-from-path",
      "type": "dql",
      "matcher": "matchesValue(log.source, \" /mnt/spark/logs/*\") OR matchesValue(log.source,
\\ /opt/spark/logs/*\")",
      "dql": {
        "script": "parse log.source, \"LD ' /mnt/spark/logs/' LD:k8s.pod_name LD '/spark-app.log'\""
      }
    },
    {
      "id": "resolve-spark-pod-entity",
      "type": "fieldsAdd",
      "matcher": "isNotNull(k8s.pod_name)",
      "fieldsAdd": {
        "fields": [
          {
            "name": "dt.source_entity",
            "value": "lookupEntity(CLOUD_APPLICATION_INSTANCE, k8s.pod_name)"
          }
        ]
      }
    }
  ]
},
"davis": {
  "processors": [
    {
      "matcher": "(loglevel == \"WARN\" OR loglevel == \"ERROR\") AND (matchesValue(log.source,
\\ /mnt/spark/logs/*\") OR matchesValue(log.source, \" /opt/spark/logs/*\"))",
      "davis": {
        "properties": [
          { "key": "event.type", "value": "ERROR_EVENT" },
          { "key": "event.name", "value": "Spark log {loglevel} on {k8s.pod_name}" },
          { "key": "event.description", "value": "Spark {loglevel} on {log.source}: {content}" },
          { "key": "dt.source_entity", "value": "{dt.source_entity}" }
        ]
      }
    }
  ]
}
}
}
}

```



OpenPipeline **Add fields** (**fieldsAdd**) accepts `lookupEntity(CLOUD_APPLICATION_INSTANCE, <pod-name-field>)` in the API **value** property (Settings API uses **name**, not **field**, for the target attribute). DQL processors in the Processing stage do **not** enable **fetch** or **smartscapeNodes**; do not use Grail notebook lookup syntax there. When lookup misses, `dt.source_entity` remains the OneAgent HOST. The required outcome is `dt.source_entity = CLOUD_APPLICATION_INSTANCE-...` before the Davis processor runs.

2.5.6. Davis processor properties

Property	Value
Matcher	(loglevel == "WARN" OR loglevel == "ERROR") AND matchesValue(log.source, "/var/log/myapp/*")

Property	Value
event.type	ERROR_EVENT
event.name	Spark log {loglevel} on {k8s.pod_name}
event.description	Application {loglevel} on {log.source}: {content}
dt.source_entity	CLOUD_APPLICATION_INSTANCE entity when pod resolved; else pass-through process or host
dt.davis.event_timeout	15m (adjust to problem correlation needs)

2.5.7. Example Grail log record at Davis input

Attribute	Example
content	2026-06-27 14:53:46 ERROR Worker:264 - All masters are unresponsive! Giving up.
loglevel	ERROR
log.source	/var/log/myapp/myapp-worker-7d4f9b-xk2lm/application.log
workload.pod_name	myapp-worker-7d4f9b-xk2lm (after fields extract)
dt.source_entity	CLOUD_APPLICATION_INSTANCE-A1B2C3D4E5F67890
host.name	k8s-node-01 (tail host — not the alert CI when pod entity is set)

2.5.8. Alerting profile

Scope the profile to the integration management zone and include ERROR and CUSTOM_ALERT severity levels (Davis log events map to ERROR):

```
{
  "schemaId": "builtin:alerting.profile",
  "scope": "environment",
  "value": {
    "name": "Application Observability - ServiceNow",
    "severityRules": [
      { "severityLevel": "AVAILABILITY", "delayInMinutes": 0, "tagFilterIncludeMode": "NONE" },
      { "severityLevel": "ERRORS", "delayInMinutes": 0, "tagFilterIncludeMode": "NONE" },
      { "severityLevel": "PERFORMANCE", "delayInMinutes": 0, "tagFilterIncludeMode": "NONE" },
      { "severityLevel": "RESOURCE_CONTENTION", "delayInMinutes": 0, "tagFilterIncludeMode": "NONE" },
      { "severityLevel": "CUSTOM_ALERT", "delayInMinutes": 0, "tagFilterIncludeMode": "NONE" }
    ],
    "eventFilters": [],
    "managementZone": "<management-zone-object-id>"
  }
}
```

2.5.9. Host CPU metric event (optional companion path)

Host CPU problems are **not Spark-specific** unless scoped. Use a **neutral event title** and an **entity filter** so the detector evaluates only hosts in the integration boundary (management zone or named hosts). Different business units may use different thresholds — create **separate metric events** per scope rather than one tenant-wide rule.

```
{
  "schemaId": "builtin:anomaly-detection.metric-events",
  "scope": "environment",
  "value": {
    "enabled": true,
    "summary": "Spark Lab - Host CPU above 80%",
    "queryDefinition": {
      "type": "METRIC_KEY",
      "metricKey": "builtin:host.cpu.usage",
      "aggregation": "AVG",
      "entityFilter": {
        "conditions": [
          {
            "type": "HOST_GROUP_NAME",
            "operator": "EQUALS",
            "value": "<integration-host-group>"
          }
        ]
      },
      "dimensionFilter": []
    },
    "eventEntityDimensionKey": "dt.entity.host",
    "modelProperties": {
      "type": "STATIC_THRESHOLD",
      "threshold": 80.0,
      "alertOnNoData": false,
      "alertCondition": "ABOVE",
      "violatingSamples": 1,
      "samples": 3,
      "dealertingSamples": 3
    },
    "eventTemplate": {
      "title": "Host ({dims:dt.entity.host}) CPU above 80%",
      "description": "Host CPU usage in management zone <name> exceeded 80%.",
      "eventType": "CUSTOM_ALERT",
      "davisMerge": false,
      "metadata": []
    }
  }
}
```

Scope the detector with an **entity filter**: **HOST_GROUP_NAME** (when OneAgent assigns a host group), **MANAGEMENT_ZONE** (numeric zone id), or individual **HOST_NAME** values. Multiple conditions in one filter use **AND** logic — use a single host group or zone rather than listing every host unless each host has its own metric event.

When the integration uses a **management zone**, a **MANAGEMENT_ZONE** filter requires the zone's **numeric id** (not the Settings API `objectId`). Template: [observability/dynatrace/tenants/<tenant>/integrations/spark-cpu-metric-event.json.j2](https://observability.dynatrace.com/tenants/<tenant>/integrations/spark-cpu-metric-event.json.j2).

2.5.10. ServiceNow (Step 2)

No direct ServiceNow configuration. Ensure SGC scheduled imports cover **CLOUD_APPLICATION_INSTANCE** and **PROCESS_GROUP** entity types that Davis may select as `dt.source_entity`.

2.6. Step 3 — Dynatrace sends the problem to ServiceNow

2.6.1. Webhook endpoint

```
POST https://<instance>.service-now.com/api/sn_em_connector/em/inbound_event?source=SG0-Dynatrace
Authorization: Basic <base64(user:password)>
Content-Type: application/json
```

Use **ProblemDetailsJSON** (v1) in the payload template for SGC — not ProblemDetailsJSONv2 (legacy templates may return HTTP 500 on some instances).

2.6.2. Problem notification payload template

The template in [Problem notification payload specification](#) is substituted by Dynatrace at send time. **ProblemDetailsJSON.rankedEvents[]** carries the OpenPipeline Davis properties from [OpenPipeline specification](#) (**event.description**, entity ids, severity). Those ranked events are the **problem specification** that flows into ServiceNow **em_event.description** (via **ProblemDetailsText**) and into **em_event.additional_info**.

Specification 2.6.2-1. Problem notification payload (SGC)

```
{
  "ConnectionId": "<sgc-connection-sys-id>",
  "ProblemID": "{ProblemID}",
  "ProblemTitle": "{ProblemTitle}",
  "ProblemSeverity": "{ProblemSeverity}",
  "ProblemImpact": "{ProblemImpact}",
  "ProblemDetailsJSON": {ProblemDetailsJSON},
  "ProblemDetailsText": "{ProblemDetailsText}",
  "ImpactedEntities": {ImpactedEntities},
  "ImpactedEntity": "{ImpactedEntity}",
  "ProblemURL": "{ProblemURL}",
  "State": "{State}",
  "Tags": "{Tags}",
  "PID": "{PID}"
}
```

Enable **notifyClosedProblems** so RESOLVED posts update the same em_event via message_key.

2.6.3. Webhook field mapping

Webhook field	Dynatrace source	ServiceNow target
ProblemID	Stable problem identifier	em_event.message_key (OPEN and RESOLVED deduplication)
ProblemTitle	event.name / Davis title	em_event.message; contributes to description
ProblemSeverity	ERROR, CUSTOM_ALERT, etc.	em_event.severity (numeric mapping per SGC rules)

Webhook field	Dynatrace source	ServiceNow target
ImpactedEntities[].entityId	CLOUD_APPLICATION_INSTANCE-..., PROCESS_GROUP-..., HOST-...	sys_object_source lookup → em_event.cmdb_ci
ProblemDetailsJSON.rankEvents[]	Ranked events with entityId, eventType	Fallback CI binding parse path
ProblemDetailsText	Davis narrative including event.description	em_event.description (primary operator-facing text)
Tags	Problem tags (Project, Environment, ...)	em_event.additional_info; not CSDM join keys
State	OPEN or RESOLVED	Create vs close/update event
ProblemURL	Deep link to Dynatrace problem	em_event.additional_info

2.6.4. Complete example webhook body

After Dynatrace substitutes placeholders for an ERROR log problem:

```
{
  "ConnectionId": "a1b2c3d4e5f678901234567890abcdef0",
  "ProblemID": "-9876543210987654321",
  "ProblemTitle": "Application log ERROR",
  "ProblemSeverity": "ERROR",
  "ProblemImpact": "INFRASTRUCTURE",
  "ProblemDetailsJSON": {
    "displayName": "Application log ERROR",
    "problemId": "-9876543210987654321",
    "rankedEvents": [
      {
        "entityId": "CLOUD_APPLICATION_INSTANCE-A1B2C3D4E5F67890",
        "entityName": "myapp-worker-7d4f9b-xk2lm",
        "eventType": "ERROR_EVENT",
        "severityLevel": "ERROR",
        "title": "Application log ERROR",
        "eventDescription": "Application ERROR on /var/log/myapp/myapp-worker-7d4f9b-xk2lm/application.log: 2026-06-27 14:53:46 ERROR Worker:264 - All masters are unresponsive! Giving up."
      }
    ],
    "startTime": 1719502426000,
    "endTime": -1
  },
  "ProblemDetailsText": "Application ERROR on /var/log/myapp/myapp-worker-7d4f9b-xk2lm/application.log:\n2026-06-27 14:53:46 ERROR Worker:264 - All masters are unresponsive! Giving up.\n\n1 impacted entity: myapp-worker-7d4f9b-xk2lm",
  "ImpactedEntities": [
    {
      "entityId": "CLOUD_APPLICATION_INSTANCE-A1B2C3D4E5F67890",
      "name": "myapp-worker-7d4f9b-xk2lm",
      "type": "CLOUD_APPLICATION_INSTANCE"
    }
  ],
  "ImpactedEntity": "myapp-worker-7d4f9b-xk2lm",
  "ProblemURL": "https://<tenant>.apps.dynatrace.com/ui/problems/<problem-id>",
  "State": "OPEN",
  "Tags": "Environment:production,Project:myapp",
  "PID": "-9876543210987654321"
}
```

```
}
```

2.6.5. Ensuring log fields reach ServiceNow

Layer	Configuration
OpenPipeline Davis processor	Set event.description to include {content} , {loglevel} , and {log.source} (Step 2); set event.name to include {k8s.pod_name} for per-pod correlation (OpenPipeline specification)
Dynatrace problem notification	When OpenPipeline Davis properties are present, Dynatrace expands them into ProblemDetailsJSON.rankedEvents[] (each ranked event includes eventDescription , entityId , title). Dynatrace then assembles ProblemDetailsText from those ranked events (including event.description text) before the webhook is sent. See Problem notification payload specification .
SGC field mapping (sa_event_field_mapping in sn_dynatrace_integ)	Map ProblemDetailsText → em_event.description ; map ImpactedEntity or parsed pod name → em_event.resource
em_event business rule (Spark pod bind)	When path or node encodes pod name, set cmdb_ci , resource , message_key prefix K8sLog-{pod}- , and message Event Log WARN or Event Log ERROR
em_event.additional_info	Persist full ProblemDetailsJSON for Event Management rules and incident scripts
em_alert.description	Event Management alert rule copies em_event.description ; cmdb_ci copied from event



Kubernetes labels (servicenow.io/application-service-identifier) are **not** in the webhook payload. They remain on CMDB pod CIs for Step 5 incident correlation.

2.6.6. ServiceNow (Step 3)

The **Observability engineer** (ServiceNow administrator with Event Management and SGC access) must verify the push connector — not the monitored application developer:

1. Confirm the Store app **Service Graph Connector for Observability – Dynatrace** (**sn_dynatrace_integ**) is installed and active.
2. In **Event Management** → **Integrations** → **Push Connectors** (or **Connections** → **Observability** → **Dynatrace** in newer releases), locate the connector whose inbound URL ends with:

POST /api/sn_em_connector/em/inbound_event?source=SGO-Dynatrace
3. Verify **Basic authentication** credentials match the Dynatrace problem notification profile (brooks-lab: provisioned via Ansible **ensure_sn_dt_webhook_auth.yml**).
4. In the scoped app, open **Event field mappings** (**sa_event_field_mapping** / **\$sa_event_map**) and confirm: **ProblemDetailsText** → **description**; **ProblemTitle** → **message**; **ProblemID** → **message_key**; severity mapping for ERROR and WARN.

5. Ensure EM **event rules** mark inbound SGO-Dynatrace events **Ready** when `cmdb_ci` is populated (see [Event Management rules](#)).
6. Ensure EM **alert rules** create alerts from Ready events and copy `cmdb_ci`, `description`, `resource`, and `node` from the event.

No **ServiceNow** configuration is required on the monitored application itself. No **Dynatrace** changes are required at this step beyond the problem notification profile deployed in [Step 2 — Dynatrace](#).

2.7. Step 4 — ServiceNow receives the event and binds infrastructure CI

2.7.1. Event listener

Setting	Value
Source	SGO-Dynatrace (URL query parameter on inbound_event)
Scoped app	sn_dynatrace_integ
CI lookup	sys_object_source where name=SGO-Dynatrace and id=<entityId from ImpactedEntities>

2.7.2. em_event field mapping

em_event field	Population
source	SGO-Dynatrace (from URL)
message_key	ProblemID
description	ProblemDetailsText — must include full log line from OpenPipeline event.description
message	ProblemTitle
severity	Mapped from ProblemSeverity (ERROR → 2 Major typical; WARN → 3 Minor if applicable)
type	ERROR or derived from ProblemDetailsJSON ranked event type
node	ImpactedEntity display name (pod name when CLOUD_APPLICATION_INSTANCE)
resource	Log file path or workload.pod_name parsed from ProblemDetailsText or additional_info
cmdb_ci	sys_object_source.target_sys_id for entityId in ImpactedEntities
cmdb_ci_type	cmdb_ci_kubernetes_pod when entity is CLOUD_APPLICATION_INSTANCE; cmdb_ci_appl for PROCESS_GROUP; cmdb_ci_linux_server for HOST fallback
additional_info	JSON: ProblemURL, Tags, ImpactedEntities, full ProblemDetailsJSON, PID

em_event field	Population
time_of_event	Problem start time from ProblemDetailsJSON
correlation_id	PID when mapped

2.7.3. CI binding procedure

1. Listener receives POST at inbound_event?source=SGO-Dynatrace.
2. Parser extracts entityId from ImpactedEntities[0] or ProblemDetailsJSON.rankedEvents[0].
3. Query sys_object_source: name=SGO-Dynatrace, id=<entityId>.
4. Set em_event.cmdb_ci and em_event.cmdb_ci_type from the matched row.

2.7.4. Entity type to CMDB class

Dynatrace entityId prefix	CMDB class	When to expect
CLOUD_APPLICATION_INSTANCE	cmdb_ci_kubernetes_pod	OpenPipeline resolved pod entity (preferred for K8s log paths)
PROCESS_GROUP	cmdb_ci_appl	Log tied to JVM process when pod lookup unavailable
HOST	cmdb_ci_linux_server	Last resort — file tailed on node without pod enrichment

Prerequisites for pod binding:

- SGC Kubernetes Pod import includes pods in the application namespace.
- Pod name in Dynatrace matches Kubernetes metadata.name.
- IRE merge aligns KVA-discovered pod CIs with SGC-imported entities when both exist.
- sys_object_source row exists before the webhook arrives (account for import schedule lag).

2.7.5. Description and alert text

Field	Required content
em_event.description	Line 1: log level and message body from content; Line 2: log.source path; optional ProblemURL appended
em_alert.description	Same as em_event.description (copied by alert rule or inherited on alert create)
em_alert.resource	Pod name or full log file path
em_alert.cmdb_ci	Same infrastructure CI as em_event.cmdb_ci

2.7.6. SGC and Event Management configuration

Component	Specification
sa_event_field_mapping	ProblemDetailsText → description; ProblemTitle → message; ProblemID → message_key; severity mapping for ERROR and WARN
EM event rule	Mark events Ready when severity and cmdb_ci are populated; optional script normalizes description from additional_info if connector mapping is insufficient
EM alert rule	Create alert from Ready events; copy cmdb_ci and description from event; require cmdb_ci for incident automation
ACL on cmdb_key_value	Roles that run incident scripts must read tags on pod CIs (see install documentation for tag ACL requirements)

2.7.7. Application and Dynatrace (Step 4)

At Step 4 the **Observability engineer** typically performs **verification**, not new Dynatrace or application configuration:

1. After a test problem fires, confirm `em_event.source=SGO-Dynatrace` and `em_event.cmdb_ci` is populated (pod CI when OpenPipeline entity lookup succeeded).
2. If `cmdb_ci` is a Linux server instead of a pod, return to [Step 2 — Dynatrace](#) and verify `k8s.pod_name` extraction and `lookupEntity(CLOUD_APPLICATION_INSTANCE, k8s.pod_name)` in the OpenPipeline pipeline ([OpenPipeline specification](#)).
3. Confirm `sys_object_source` contains a row with `name=SGO-Dynatrace` and `id=<entityId>` from the webhook **before** the event arrives (SGC scheduled import lag is a common root cause of empty CI).
4. Deploy or verify the **K8s log pod bind** business rules ([K8s log pod bind business rules](#)) as a fallback when the webhook still reports a HOST entity but the log path encodes a pod name.

No ServiceNow-side changes are required on the monitored **application** (the Java/Spark/Python workload). **No additional Dynatrace** configuration is required at Step 4 if Steps [Step 2 — Dynatrace](#) and Step 3 are correct.

ServiceNow cannot bind to `cmdb_ci_kubernetes_pod` unless [Step 2 — Dynatrace](#) sets `dt.source_entity` to the pod entity and Step 0 imported that entity into `sys_object_source`.

2.8. Step 5 — Create incident bound to Application Service

2.8.1. Severity policy

Environment	Automatic incident when
Non-production / test	ERROR and WARN (simplifies validation when ERROR signals are hard to trigger)

Environment	Automatic incident when
Production	ERROR only; WARN remains at alert level or creates a lower-priority task per ITSM policy

Incidents must set **incident.cmdb_ci** to **cmdb_ci_service_discovered** (Application Service), not the infrastructure CI on the alert.

2.8.2. Layer model

Layer	Table	cmdb_ci class
Event	em_event	Infrastructure: cmdb_ci_kubernetes_pod, cmdb_ci_appl, or cmdb_ci_linux_server
Alert	em_alert	Same infrastructure CI as the source event
Incident	incident	cmdb_ci_service_discovered (Application Service)

2.8.3. Incident correlation

Incident correlation defines how enriched alerts map to **incident.cmdb_ci**. The business rule **em-alert-create-k8s-log-incident** tries the **Client-side** log path first, then the **Service-side** Contains traversal.

The comparison table summarizes alert CI versus incident CI for each pattern. Full algorithms follow in the adjacent subsections.

Pattern	Alert CI	incident.cmdb_ci resolution
Service-side	cmdb_ci_kubernetes_pod (preferred) or HOST fallback	Spark Service-Side Log Pattern — Contains traversal from pod CI
Client-side	HOST or empty (acceptable)	Spark Client-Side Log Pattern — /logs/spark-client/ → Spark Client AS

2.8.3.1. Service-side

The **Spark Service-Side Log Pattern** resolves Application Services by traversing **Contains::Contained by** from the alert's pod CI. A materialized edge from **cmdb_ci_service_discovered** (parent) to **cmdb_ci_kubernetes_pod** (child) **must** exist before the problem fires.



Figure 4. Incident correlation — Spark Service-Side Log Pattern

1. **K8sLogPodCiBind** sets **em_event.cmdb_ci** and **em_alert.cmdb_ci** to the **pod CI** whose **name** matches the log path segment (Step 4).
2. Query **cmdb_rel_ci** where **child** = pod CI sys_id and **type** = **Contains::Contained by**.

3. Parent row **must** be `cmdb_ci_service_discovered`.
4. Set `incident.cmdb_ci` = parent Application Service `sys_id`.

Out of scope: host CI binding without pod rebinding, process group CI, tag-only lookup without a Contains edge.

2.8.3.2. Client-side

The **Spark Client-Side Log Pattern** maps log paths to a single logical Application Service. Client-mode driver logs **do not** correspond to a `cmdb_ci_kubernetes_pod`; incident automation **must not** depend on alert CI or Contains edges.

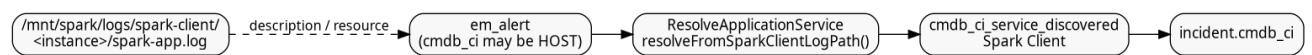


Figure 5. Incident correlation — Spark Client-Side Log Pattern

1. Scan `em_alert.description`, `resource`, and `node` for the substring `/logs/spark-client/`.
2. Look up Application Service **Spark Client** by **name** (see [CSDM identifier lookup](#)).
3. Set `incident.cmdb_ci` to that Application Service `sys_id`.
4. `K8sLogPodCiBind` skips path segment `spark-client` when parsing pod names.

2.8.4. CSDM identifier lookup

CSDM **identifier** in `*.csdm.yaml` is the logical machine key embedded in log paths and optional Dynatrace tags. The table below shows which CMDB fields are reliable for incident lookup on `optimizincdemo1`.

Purpose	Mechanism	Queryable field on <code>cmdb_ci_service_discovered</code>
Manual / logical AS documentation	CSDM identifier : <code>spark-client</code> in log path contract	identifier column — often not populated on <code>optimizincdemo1</code>
Display / incident lookup	CSDM name (e.g. <code>Spark Client</code>)	name — reliable for <code>resolveFromSparkClientLogPath()</code>
Optional process tag	<code>DT_TAGS=servicenow.io/application-service-identifier=spark-client</code>	Process-group problems only; not required for log-path incidents

Spark Client AS **must** have `service_status=operational` (promoted by `csdm/deploy.yml` → `ensure_host_as_operational.yml`).

```

cd ansible
ansible-playbook -i inventory.yml playbooks/servicenow/incident/verify_spark_client_as.yml \
  -e @../vars/secrets.yaml
  
```

2.8.5. Known limitation — Davis problem bundling

Dynatrace Davis **bundles** multiple log-derived `ERROR_EVENT` rows that share a root-cause analysis into **one problem**. When client-mode and Service-side log lines fire within the same Davis

correlation window, a **single** `em_alert.description` can contain **both**:

- Application log WARN on spark-client with path /mnt/spark/logs/spark-client/...
- Application log WARN on spark-master-0 with path /mnt/spark/logs/spark-master-0/...

How this happens:

1. OpenPipeline emits separate Davis events per log line with distinct `event.name` values (... on spark-client vs ... on spark-master-0).
2. Davis root-cause analysis treats concurrent chapter activity on the same host/cluster as **one problem** when errors correlate temporally.
3. The ServiceNow webhook receives **one** problem payload whose description concatenates **all** bundled event summaries.
4. `em-alert-create-k8s-log-incident` scans for /logs/spark-client/ **first**; if absent from the bundled text ordering, **Service-side** correlation may win and set `incident.cmdb_ci` to **Spark Master** even when the operator intended **Spark Client**.
5. Event Management may also pre-link alerts to an existing open incident (`INC0013763`) before the business rule corrects Application Service assignment.

Impact: Client-side and Service-side validation in the same test window can produce **misleading incident CI** on bundled alerts. Split test runs by path pattern or wait for Davis timeout (lab: `15m dt.davis.event_timeout`) between pattern tests until remediation is implemented.

2.8.6. ServiceNow incident automation specification

Component	Specification
Script Include: <code>ResolveApplicationService</code>	Service-Side: <code>cmdb_rel_ci</code> Contains::Contained by from pod CI to Application Service. Spark client: <code>resolveFromSparkClientLogPath()</code> when alert text contains /logs/spark-client/ → Application Service Spark Client
Script Include: <code>K8sLogPodCiBind</code>	Rebind <code>em_event</code> / <code>em_alert</code> from initial SGO HOST binding to <code>cmdb_ci_kubernetes_pod</code> when log path segment matches a discovered pod name; skip segment <code>spark-client</code> ; propagate binding to alerts sharing <code>message_key</code>
EM event rule	Mark Spark log events Ready when <code>source=SG0-Dynatrace</code> , severity matches policy, and <code>cmdb_ci</code> is set
EM alert rule	Create alert from Ready events; copy <code>cmdb_ci</code> , <code>description</code> , <code>resource</code> , <code>node</code> from event
Business rule: <code>em-event-bind-k8s-log-pod-ci</code>	Before insert/update on <code>em_event</code> ; order 5000 ; enriches <code>cmdb_ci</code> , <code>resource</code> , <code>message_key</code> via <code>K8sLogPodCiBind</code> (must run before EM Ready/alert rules persist the row)
Business rule: <code>em-event-propagate-k8s-log-pod-ci</code>	After insert/update on <code>em_event</code> ; order 5005 ; copies pod binding from event to related <code>em_alert</code> rows sharing <code>message_key</code> (runs after the event row is saved)

Component	Specification
Business rule: <code>em-alert-bind-k8s-log-pod-ci</code>	Before insert/update on <code>em_alert</code> ; order 5010 ; same pod/path enrichment when alert is created or updated directly (covers SGC webhook path that skips a separate bind on the event)
Business rule: <code>em-alert-create-k8s-log-incident</code>	After insert/update on <code>em_alert</code> ; order 5020 ; tries spark-client path first, else Service-Side Contains; creates or correlates incident, sets <code>em_alert.incident</code>
Severity filter	Non-prod: <code>em_alert.severity ≤ 3</code> (ERROR and WARN); Production: <code>em_alert.severity = 2</code> (ERROR only)
Duplicate suppression	Correlate open incidents by Application Service + short description prefix <code>Event Log</code> ; use <code>ProblemID / message_key</code> for event lifecycle

2.8.7. Business rule order and rationale

Business rules on `em_event` and `em_alert` run in **order** within the same **when** phase (**before** or **after**). Lower order runs first.

Order	Rule	Why this position
5000	<code>em-event-bind-k8s-log-pod-ci</code> (before)	Parse log path and set <code>cmdb_ci / resource / message_key</code> before the row is written. EM event rules that mark the event Ready and copy fields to alerts depend on these values being present on insert.
5005	<code>em-event-propagate-k8s-log-pod-ci</code> (after)	After the event is saved, push binding to sibling <code>em_alert</code> rows. Must be after (not before) because it updates other records.
5010	<code>em-alert-bind-k8s-log-pod-ci</code> (before)	Same enrichment for alerts created or updated by the SGC webhook / EM alert rule without going through a separate event bind cycle.
5020	<code>em-alert-create-k8s-log-incident</code> (before)	Needs enriched <code>resource</code> and log-path context from 5000/5010 in the same transaction. Creates the incident and sets <code>em_alert.incident</code> before the alert row is committed. Using after + <code>current.update()</code> was unreliable for REST-driven updates.

2.8.8. Alert to incident linkage (`em_alert.incident`)

On `optimizincdemo1`, Event Management links an alert to its incident through `em_alert.incident` (reference to the incident record). That field populates the incident **Alerts** tab. Some instances expose the same reference as `em_alert.task`; this build uses **incident only** — do not reference `task`.

Creating an incident with `GlideRecord('incident').insert()` alone does **not** set this link — automation must assign `current.incident = <incident_sys_id>` on the alert in the **before** business rule.

Incident short description format: **Event Log WARN** or **Event Log ERROR** (parsed from alert description).

2.8.9. Problem and alert lifecycle (OPEN / RESOLVED)

Spark/Log4j does **not** emit explicit “clear” or “recovery” log lines for this use case. Closure is driven by **Dynatrace problem lifecycle**, not application log content:

1. OneAgent tails application log files on the **host node** (see [Log emission](#)).
2. OpenPipeline Davis turns qualifying **ERROR** / **WARN** lines into Davis events; repeated lines merge into one **problem** (`dt.davis.event_timeout: 15m`).
3. When matching log activity stops, Dynatrace **auto-closes** the problem after the timeout.
4. The brooks-lab problem notification has `notifyClosedProblems: true`, so Dynatrace POSTs a **RESOLVED** webhook with the same **ProblemID** as the OPEN post.
5. SGO maps `message_key = ProblemID`, so ServiceNow **updates the same em_event** (does not create a duplicate).
6. Event Management then transitions the linked **alert** (and optionally the **incident**) according to configured EM rules when the event state/resolution changes.

Verify RESOLVED handling: `em_event` where `source=SGO-Dynatrace` and `resolution_code` is set after problems clear; linked `em_alert.state` should move to **Closed** when EM auto-close rules are configured.

2.8.10. OpenPipeline and custom-source attributes (generic Log4j pattern)

Attributes added by automation (not Spark-specific semantics):

Attribute	Source	Purpose
<code>k8s.log.path</code>	Custom log source enrichment	Full file path of the log line
<code>k8s.pod_name</code>	Path segment after <code>/logs/</code>	Must equal <code>cmdb_ci_kubernetes_pod.name</code>
<code>event.name</code>	OpenPipeline Davis	Application log {loglevel} on {k8s.pod_name}
<code>event.description</code>	OpenPipeline Davis	Application {loglevel} on {k8s.log.path}: {content}

Pod **Kubernetes labels** (`servicenow.io/application-service-identifier`, etc.) are the CMDB/service-mapping contract — not Dynatrace tags. See [pod labels](#).



resolve-k8s-pod-entity (OpenPipeline `lookupEntity` for pod Smartscape) is **disabled** so problems stay on **HOST** entities inside the Spark Observability management zone. ServiceNow `K8sLogPodCiBind` rebinds alerts to the **pod CI** from the log path; `ResolveApplicationService` traverses the **Contains::Contained** by edge to the Application Service (Service-Side pattern only).

2.8.11. Event Management rules (manual configuration)

Configure under **Event Management** → **Rules**:

Rule	Condition and action
Event rule (K8s application logs)	When: <code>em_event</code> insert/update; Filter: <code>source=SGO-Dynatrace</code> <code>severity<=3</code> <code>cmdb_ciISNOTEMPTY</code> ; Action: set state to Ready (or use existing Ready rule scoped to SGO-Dynatrace)
Alert rule (K8s application logs)	When: Ready event; Filter: <code>source=SGO-Dynatrace</code> ; Action: create/update alert; copy <code>cmdb_ci</code> , <code>description</code> , <code>resource</code> , <code>node</code> , <code>message_key</code> from event

Scope Spark-specific business rules with `source=SGO-Dynatrace` so legacy `source=Dynatrace` events are unaffected during connector migration (see [Legacy and SGO-Dynatrace coexistence](#)).

2.8.12. Script Includes

Deploy from `servicenow/integrations/incident/`. Global scope, **Active** = true. Ansible: `ansible/playbooks/servicenow/incident/deploy.yml`.

Each Script Include below summarizes **purpose**, **inputs**, **outputs** / **side effects**, and how it fits the business rules. Source of truth is the `.si.js` file in the repository.

2.8.12.1. ResolveApplicationService

Purpose: Resolve `cmdb_ci_service_discovered` (Application Service) `sys_id` for incident binding.

Methods:

- `resolveFromInfrastructureCi(ciSysId)` — **Spark Service-Side Log Pattern**. Queries `cmdb_rel_ci` for **Contains::Contained by** where `child` = pod CI and `parent` is an Application Service. Returns parent `sys_id` or `null`.
- `resolveFromSparkClientLogPath(text)` — **Spark Client-Side Log Pattern**. When `text` contains `/logs/spark-client/`, looks up Application Service **Spark Client** by `identifier=spark-client`, then fallback `name=Spark Client`.

Inputs: Workload CI `sys_id` (Service-side) or alert description/resource text (Client-side).

Outputs: Application Service `sys_id` for `incident.cmdb_ci`; no CMDB writes.

Side effects: None (read-only GlideRecord queries).

Source: `servicenow/integrations/incident/ResolveApplicationService.si.js`

2.8.12.2. K8sLogPodCiBind

Purpose: Rebind `em_event` and `em_alert` from HOST or empty CI to `cmdb_ci_kubernetes_pod` when the log path directory segment matches a discovered pod name (OpenPipeline often leaves problems on HOST).

Methods:

- `resolvePodName(description, resource, node)` — Scans concatenated fields for

/.../logs/<segment>; skips `spark-client` and wildcards; verifies pod exists in CMDB.

- `applyPodBinding(gr)` — Sets `cmdb_ci`, `node`, `resource`, `message_key` prefix `K8sLog-{pod}-`, and `message` to Event Log WARN/ERROR when generic. Returns `true` on success. Caller must `update()`.
- `propagateEventToAlerts(eventGr)` — Copies pod binding from event to related alerts sharing `message_key`.

Inputs: `em_event` or `em_alert` GlideRecord.

Outputs: Mutated fields on current row; optional `em_alert.update()` on related rows.

Source: `servicenow/integrations/incident/K8sLogPodCiBind.si.js`

2.8.13. Business rules (prescriptive configuration)

Create under **System Definition** → **Business Rules**. All three run **After** insert and update. Filter conditions must include `source=SG0-Dynatrace` so legacy connectors are not modified.

Name	Table	Order	Filter condition
<code>em-event-bind-k8s-log-pod-ci</code>	<code>em_event</code>	5000	<code>source=SG0-Dynatrace</code>
<code>em-alert-bind-k8s-log-pod-ci</code>	<code>em_alert</code>	5010	<code>source=SG0-Dynatrace</code>
<code>em-alert-create-k8s-log-incident</code>	<code>em_alert</code>	5100	<code>source=SG0-Dynatrace^severity<=3</code>

2.8.13.1. em-event-bind-k8s-log-pod-ci

```
(function executeRule(current, previous /*null on insert*/) {
  var binder = new K8sLogPodCiBind();
  if (!binder.applyPodBinding(current)) {
    return;
  }

  current.update();
  binder.propagateEventToAlerts(current);
})(current, previous);
```

2.8.13.2. em-alert-bind-k8s-log-pod-ci

```
(function executeRule(current, previous /*null on insert*/) {
  var binder = new K8sLogPodCiBind();
  if (!binder.applyPodBinding(current)) {
    return;
  }

  current.update();
})(current, previous);
```

2.8.13.3. em-alert-create-k8s-log-incident

Creates or correlates an incident on the Application Service, then sets `current.task` so the alert

appears on the incident **Alerts** tab.

```
(function executeRule(current, previous /*null on insert*/) {
  if (current.source.toString() !== 'SGO-Dynatrace') {
    return;
  }

  if (!current.task.nil()) {
    return;
  }

  var severity = parseInt(current.severity, 10);
  if (isNaN(severity) || severity > 3) {
    return;
  }

  var desc = current.description.toString();
  if (
    desc.indexOf('Spark ') === -1 &&
    desc.indexOf('spark-app.log') === -1 &&
    desc.indexOf('/mnt/spark/logs/') === -1
  ) {
    return;
  }

  if (current.cmdb_ci.nil()) {
    return;
  }

  var resolver = new ResolveApplicationService();
  var asSysId = resolver.resolveFromInfrastructureCi(current.cmdb_ci.toString());
  if (!asSysId) {
    return;
  }

  var levelMatch = desc.match(/\b(ERROR|WARN)\b/);
  var logLevel = levelMatch
    ? levelMatch[1]
    : severity <= 2
      ? 'ERROR'
      : 'WARN';
  var shortDesc = 'Event Log ' + logLevel;
  var shortDescPrefix = 'Event Log';

  function linkAlertToIncident(incSysId) {
    var alertGr = new GlideRecord('em_alert');
    if (!alertGr.get(current.sys_id)) {
      return;
    }
    alertGr.task = incSysId;
    alertGr.setWorkflow(false);
    alertGr.update();
  }

  var backfillInc = new GlideRecord('incident');
  backfillInc.addQuery('active', true);
  backfillInc.addQuery('short_description', 'STARTSWITH', shortDescPrefix);
  backfillInc.addEncodedQuery('cmdb_ciISEMPTY');
  backfillInc.orderBy('sys_created_on');
  backfillInc.setLimit(1);
  backfillInc.query();
  if (backfillInc.next()) {
    backfillInc.cmdb_ci = asSysId;
    backfillInc.short_description = shortDesc;
    backfillInc.work_notes =
      'Set Configuration item from alert ' +
      current.number +

```

```

    ' (Application Service resolved from pod CI).';
    backfillInc.update();
    linkAlertToIncident(backfillInc.sys_id);
    return;
}

var openInc = new GlideRecord('incident');
openInc.addQuery('cmdb_ci', asSysId);
openInc.addQuery('active', true);
openInc.addQuery('short_description', 'STARTSWITH', shortDescPrefix);
openInc.setLimit(1);
openInc.query();
if (openInc.next()) {
    openInc.work_notes =
        'Correlated alert ' +
        current.number +
        ': ' +
        current.description.toString().substring(0, 500);
    openInc.update();
    linkAlertToIncident(openInc.sys_id);
    return;
}

var inc = new GlideRecord('incident');
inc.initialize();
inc.cmdb_ci = asSysId;
inc.short_description = shortDesc;
inc.description = current.description;
inc.severity = current.severity;
inc.impact = 2;
inc.urgency = 2;
inc.caller_id = gs.getUserID();
inc.comments = 'Auto-created from Event Management alert ' + current.number;
inc.work_notes = 'Source alert: ' + current.number;
var incSysId = inc.insert();
linkAlertToIncident(incSysId);
})(current, previous);

```

2.8.14. Ansible deployment

```

cd ansible
ansible-playbook -i inventory.yml playbooks/servicenow/incident/deploy.yml \
-e @../vars/secrets.yaml

```

Playbook path: **ansible/playbooks/servicenow/incident/deploy.yml**. Source files: **servicenow/integrations/incident/*.js**.

2.8.15. Example resolution (Kubernetes worker pod)

Step	Record / relationship
Problem	Davis problem; impacted entity CLOUD_APPLICATION_INSTANCE-A1B2C3D4E5F67890
Event	em_event; cmdb_ci → cmdb_ci_kubernetes_pod myapp-worker-7d4f9b-xk2lm
Alert	em_alert; same pod CI; description contains full ERROR log line
Tag on pod CI	cmdb_key_value: servicenow.io/application-service-identifier=myapp-worker-zone-a

Step	Record / relationship
SM edge	Application Service myapp-worker-zone-a Contains pod CI
Incident	incident.cmdb_ci → Application Service myapp-worker-zone-a (via Service-Side Contains traversal)

2.8.16. Application and Dynatrace (Step 5)

No configuration on the application or Dynatrace side. Dynatrace problem tags (Project, Environment) do not replace CSDM tags on pod CIs for incident binding.

Chapter 3. Legacy and SGO-Dynatrace coexistence

During migration from a legacy Event Management push connector (`source=Dynatrace` on the inbound URL) to Service Graph Connector (`source=SGO-Dynatrace`), **both paths may be active**. Use `em_event.source` and `em_alert.source` to scope automation — never assume a single connector.

3.1. Separation by source

Layer	Legacy (<code>source=Dynatrace</code>)	SGC (<code>source=SGO-Dynatrace</code>)
Inbound URL	<code>/inbound_event?source=dynatrace&sys_id=<connector-sys-id></code>	<code>/inbound_event?source=SGO-Dynatrace</code>
CI binding	Legacy connector field mapping + <code>sys_object_source</code> (may differ from SGC import)	SGC <code>ImpactedEntities</code> → <code>sys_object_source</code> (name SGO-Dynatrace)
Spark log business rules	Must not run — filter <code>source=SGO-Dynatrace</code> on every BR	Pod rebind, incident create, Application Service resolution
Problem notification	Pre-existing tenant profiles (for example Demo integrations)	Dedicated profile scoped to integration management zone

3.2. Scoping rules and profiles

- **Dynatrace alerting profiles:** Scope each profile to a **management zone** (or tag filter) so only in-scope problems forward to ServiceNow. A profile attached to a legacy webhook should not receive problems from hosts outside its intended boundary.
- **ServiceNow EM rules:** Add `source=SGO-Dynatrace` (or `source=Dynatrace`) to event and alert rule filter conditions so each rule set applies only to its connector.
- **Business rules:** All Spark log automation in this document filters `source=SGO-Dynatrace`. Legacy CPU or infrastructure alerts continue through legacy rules until that connector is retired.

3.3. Host CPU alerts (not Spark-specific)

Metric events on `builtin:host.cpu.usage` fire per **host**, not per application. Even on the SGC path, a CPU problem binds `em_alert.cmdb_ci` to the breaching **host CI** — that is correct behavior.

To avoid misleading titles and out-of-scope hosts:

1. Use a **neutral title**: `Host ({dims:dt.entity.host}) CPU above 80%` (not application-branded text).
2. Add an **entity filter** on the metric event: **management zone** (preferred) or explicit **HOST_NAME** list for integration hosts.

3. Create **separate metric events** per organization or environment when thresholds differ (for example 80% in lab, 90% in production).

Template: `observability/dynatrace/tenants/<tenant>/integrations/spark-cpu-metric-event.json.j2.` Deploy via `ansible/playbooks/servicenow/sgc/sources/dynatrace/tasks/apply_dt_anomaly_detectors.yml.`

3.4. When to retire legacy connector

After SGC cutover is verified (events arrive with `source=SGO-Dynatrace`, pod CI binding works, incidents link on **Alerts** tab):

1. Disable legacy Dynatrace problem notification profiles pointing at `source=dynatrace`.
 2. Leave legacy EM rules in place until no new `source=Dynatrace` events arrive.
 3. Remove legacy rules once the connector is decommissioned.
-